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Sheet 3: NID100-5 5 V supply, Pi USB-C output, radar 5 V
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20 V inlet and eFuse



File: inlet.kicad_sch

5 V supply and outputs



File: supply_5v.kicad_sch

3.3 V, Pi link, sensors, radar, echo-ref



File: io.kicad_sch

Amplifier



File: amplifier.kicad_sch

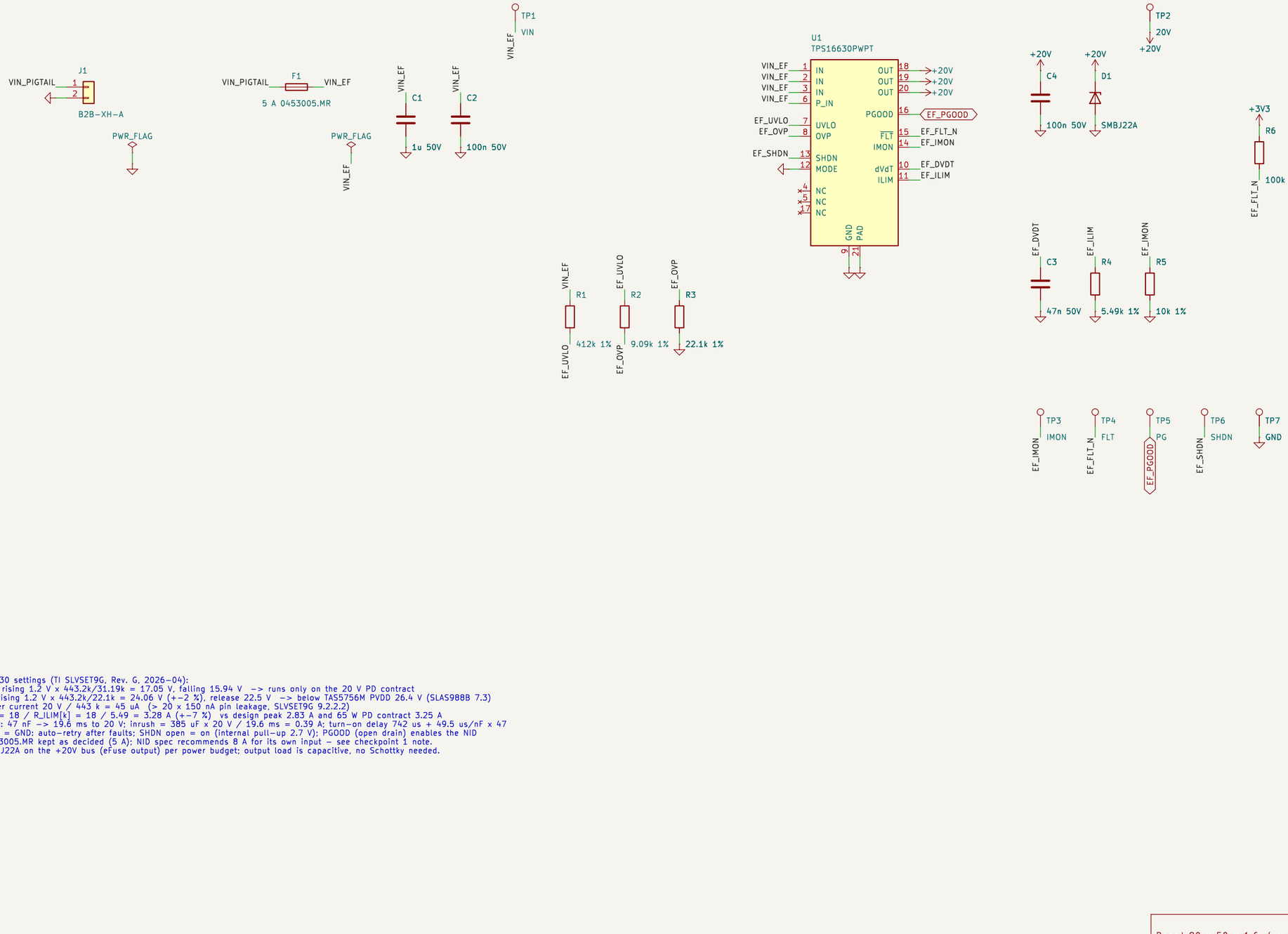
Board 90 x 50 x 1.6, 4 x Ø2.7 on 83 x 43
Brief claude/brief-electronics-design-2026-11-07.md
KiCad 9.0.9 — generated by hardware/main-board/gen (checkpoint 1)
HA panel (neocleous/ha-panel)

Sheet: /
File: main-board.kicad_sch

Title: HA panel main board

Size: A4	Date: 2026-11-07	**Rev: 1**
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20 V INLET (5807 pigtail -> fuse -> TPS16630 eFuse -> +20V bus)



TPS16630 settings (TI SLVSET9G, Rev. G, 2026-04):
  UVLO rising 1.2 V x 443.2k/31.19k = 17.05 V, falling 15.94 V -> runs only on the 20 V PD contract
  OVP rising 1.2 V x 443.2k/22.1k = 24.06 V (+-2 %), release 22.5 V -> below TAS5756M PVDD 26.4 V (SLAS988B 7.3)
  Divider current 20 V / 443 k = 45 uA (> 20 x 150 nA pin leakage, SLVSET9G 9.2.2.2)
  I_OL = 18 / R_ILIM[k] = 18 / 5.49 = 3.28 A (+-7 %) vs design peak 2.83 A and 65 W PD contract 3.25 A
  dV/dt: 47 nF -> 19.6 ms to 20 V; inrush = 385 uF x 20 V / 19.6 ms = 0.39 A; turn-on delay 742 us + 49.5 us/nF x 47
  MODE = GND: auto-retry after faults; SHDN open = on (internal pull-up 2.7 V); PGOOD (open drain) enables the NID
F1 0453005.MR kept as decided (5 A); NID spec recommends 8 A for its own input - see checkpoint 1 note.
D1 SMBJ22A on the +20V bus (eFuse output) per power budget; output load is capacitive, no Schottky needed.

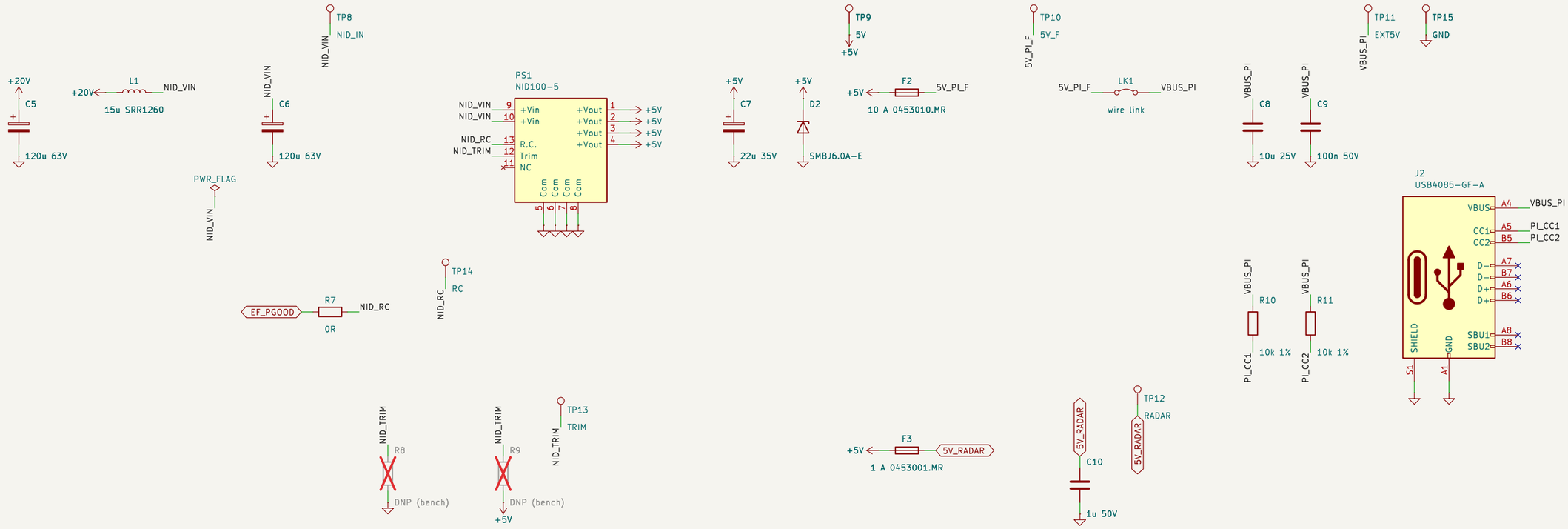
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Sheet: /20 V inlet and eFuse/
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Title: Main board - 20 V inlet and eFuse

Size: A3	Date: 2026-11-07	**Rev: 1**
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5 V SUPPLY (Mean Well EMC circuit -> NID100-5 -> Pi USB-C output, radar branch)



Mean Well NID100-SPEC (2024-10-30): EMC circuit C1/C2 120 uF/63 V, L1 15 uH (NiZn), C3 22 uF/35 V; ext. C max 100 uF/16 V;
  fuse recommended 8 A; R.C.: on = open or 1.2-12 V, off = < 0.4 V; pin 12 "Trim (optional)", no formula published.
L1 is a single winding in +Vin: the input return is the common board ground (non-isolated NID, amp on the same ground).
External C on +5V: C7 22 uF + C8 10 uF (X7R 1206, -70 % at 5 V) + C9 + C10 + LDO input path (behind 2.2 Ohm) - 35 uF << 100 uF.
PI output: 0453010.MR -> LK1 (wire link) -> J2 USB4085 VBUS; CC1/CC2 Rp 10 k to 5 V (3 A); Pi uses PSU_MAX_CURRENT=5000.
J2 USB4085-GF-A (THT, decision 3A) sits on the door side, mouth down at the bottom edge; all other parts on the Pi side.
Bench 2.7 test pads: 5V (NID out) | 5V_F (after fuse) | EXT5V (at J2) ; LK1 opened for the series current.
Trim network DNP (D23/V20): R8 Trim->Com raises Vout, R9 Trim->+Vout lowers it (FB-divider convention, not published
  by Mean Well). Spec lists pin 12 as "Trim (optional)" with no formula; on arrival measure R(pin 12-Com) unpowered and
  Vout with a trimmer clipped from TP13 to GND before fitting R8. No response at pin 12 -> alternative-regulator decision.

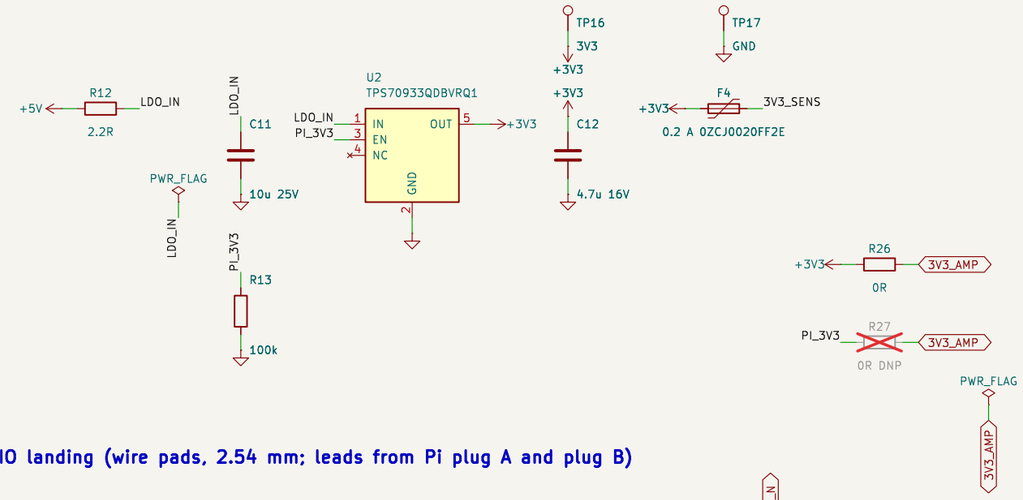
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Sheet: /5 V supply and outputs/
File: supply_5v.kicad_sch

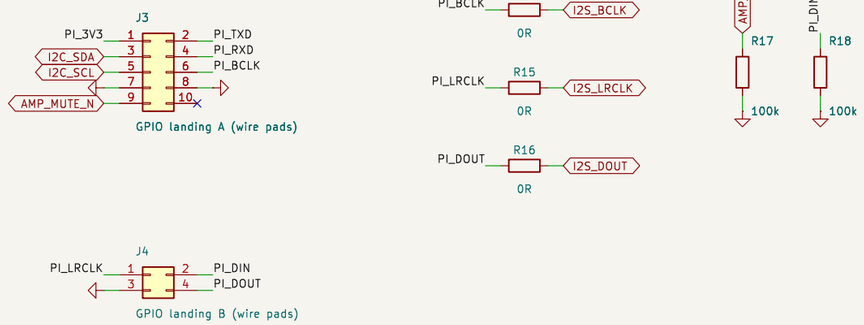
Title: Main board — 5 V supply and outputs

Size: A3	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 3/5

3.3 V LDO (follows the Pi 3V3)

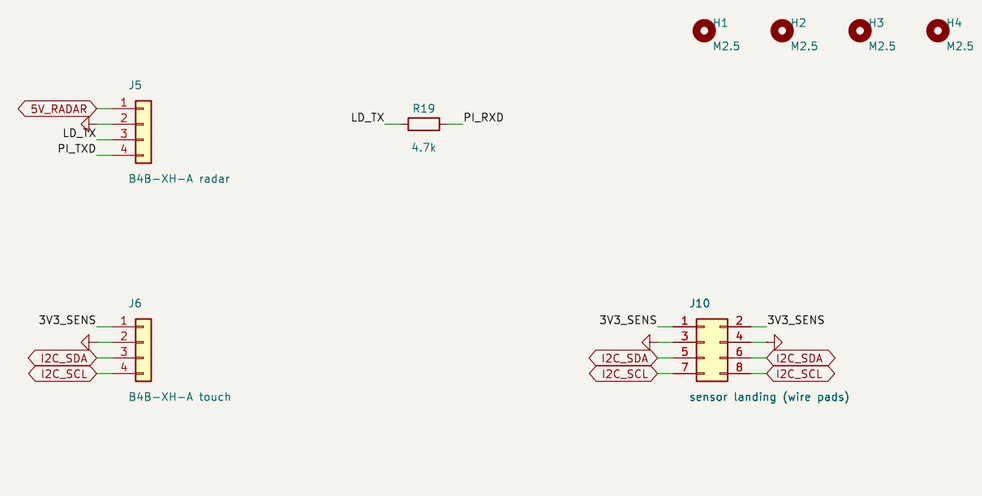


Pi GPIO landing (wire pads, 2.54 mm; leads from Pi plug A and plug B)

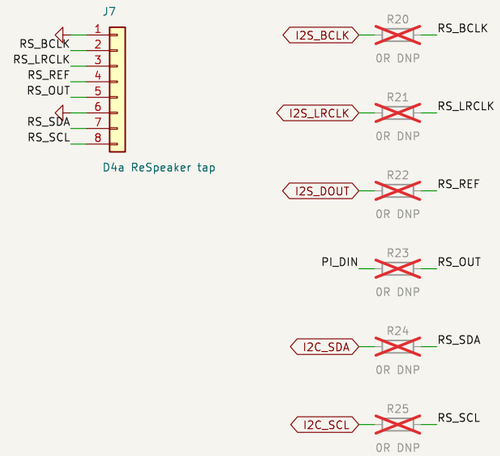


V17: Raspberry Pi publishes no current rating for the Pi 5 header 3V3 pins -> 3.3 V LDO on this board (brief item 4).
U2 TPS70933: 2.7–30 V in (survives the NID OVP 6.4–7.5 V). 150 mA, EN follows the Pi 3V3 (pin 1 lead, ~33 uA).
Loads: TAS5756M DVDD/AVDD/CPVDD 35 mA (SLAS988B 7.12) + touch/BME680/VEML6030 via F4.
STANDBY (Pi 3V3 off, 5 V on, HAT+ spec 2.2): LDO off -> no back-feed through I2C; amp muted by R17.
R26/R27: amp logic rail 3V3_AMP from the LDO (default, R26) or from the Pi 3V3 (R27): TAS5756M VIH = 0.9 x DVDD (SLAS988B 7.3) and TI advises driving its inputs from the DVDD supply; LDO +–1 % (SBVS186H) vs Pi 3V3 (not published).
Pi pins: 3/5 SDA/SCL (GPIO2/3, 1.8 k pull-ups on the Pi); 8/10 TXD/RXD (GPIO14/15, uart0):
12/35/38/40 = GPIO18/19/20/21 I2S BCLK/LRCLK/DIN/DOUT; 15 = GPIO22 amp mute (rpi-digiampplus).
I2C map: AT42QT1070 0x1B, VEML6030 0x48, TAS5756M 0x4C, BME680 0x76/0x77 (ReSpeaker 0x42/0x18 only if R24/R25 fitted).
Sensor branch 3V3_SENS (F4 0.2 A PTC): J6 touch board, J10 BME680 + VEML6030 (leads moved from the touch board, 5A).
Topology B (bench 2.2): fit R20–R23 (+R24/R25 for XU316 I2C), jumper J7 to the ReSpeaker XIAO footprint, I2S firmware.

XH headers (JST XH 2.50 mm) – radar, touch board



Echo-reference (D4a) – default topology A: all links OPEN



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Sheet: /3.3 V, Pi link, sensors, radar, echo-ref/
File: io.kicad_sch

Title: Main board – 3.3 V, Pi link, XH, D4a

Size: A3	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 4/5

TAS5756M stereo BTL (TI SLAS988B Fig. 79 / Table 22; DigiAMP+ wiring for the rpi-digiampplus overlay)

+20V → F5 (2.6 A 2920L260/33DR) → PVDD

PWR_FLAG → PVDD

TP20 — PVDD probe point: F5 pin 2 (back, 2920 land). No TP18: no free back spot on the PVDD side (layout 2026-11-07).

PVDD → GND

C100	0.1u 50V	
C101	22u 35V	
C102	22u 35V	
C121	0.1u 50V	
C122	22u 35V	
C123	22u 35V	

TP19 — FAULT → AMP_FAULT_N

R102	3V3_AMP	100k
R103	AMP_GPIO0	100k
R104	AMP_GPIO1	100k
R105	AMP_GPIO2	100k

U100 — TAS5756MDCAR

Pin	Name	Net
23	SCLK	I2S_BCLK
25	LRCK/FS	I2S_LRCLK
24	SDIN	I2S_DOUT
22	MCLK	
16	SDA	I2C_SDA
17	SCL	I2C_SCL
26	ADR0	
20	ADR1	
18	GPIO0	AMP_GPIO0
19	GPIO1	AMP_GPIO1
21	GPIO2	AMP_GPIO2
27	SPK_MUTE	AMP_MUTE_N
40	SPK_FAULT	AMP_FAULT_N
30	DVDD	3V3_AMP
31	CPVDD	3V3_AMP
14	AVDD	3V3_AMP
28	DVDD_REG	DVDD_REG
32	CP	CP
34	CN	CN
35	CPVSS	CPVSS
10, 15, 29, 33, 3, 39, 46, 49	AGND, AGND, DGND, GND, PGND, PGND, PGND, PAD	
6, 7, 41, 42, 43	PVDD	PVDD
8	GVDD_REG	GVDD
9	SPK_GAIN/FREQ	SPK_GAIN
5	BSTRPA+	BST_AP
4	SPK_OUTA+	OUT_AP
2	SPK_OUTA-	OUT_AN
1	BSTRPA-	BST_AN
44	BSTRPB+	BST_BP
45	SPK_OUTB+	OUT_BP
47	SPK_OUTB-	OUT_BN
48	BSTRPB-	BST_BN
13	DAC_OUTA	DAC_A
12	SPK_INA+	INA_P
11	SPK_INA-	INA_N
36	DAC_OUTB	DAC_B
37	SPK_INB+	INB_P
38	SPK_INB-	INB_N

GVDD → C103 (1u 50V)

GVDD → R101 (150k 1%) → SPK_GAIN

SPK_GAIN → R100 (100k 1%)

C105	3V3_AMP	1u 25V
C119	3V3_AMP	1u 25V
C118	DVDD_REG	1u 25V
C120	CPVSS	1u 25V

C104	BST_AP	OUT_AP, 0.22u 50V
C108	BST_AN	OUT_AN, 0.22u 50V
C115	BST_BP	OUT_BP, 0.22u 50V
C111	BST_BN	OUT_BN, 0.22u 50V

CP — C112 (1u 25V) — CN

C106	DAC_A	INA_P	2.2u 25V
C107	INA_N		2.2u 25V
C113	DAC_B	INB_P	2.2u 25V
C114	INB_N		2.2u 25V

L100	OUT_AP	SPK_L_P	10u
L101	OUT_AN	SPK_L_N	10u
L103	OUT_BP	SPK_R_P	10u
L102	OUT_BN	SPK_R_N	10u

C109	SPK_L_P	0.68u 50V
C110	SPK_L_N	0.68u 50V
C117	SPK_R_P	0.68u 50V
C116	SPK_R_N	0.68u 50V

J8 — B2B-XH-A speaker L: SPK_L_P (1), SPK_L_N (2)

J9 — B2B-XH-A speaker R: SPK_R_P (1), SPK_R_N (2)

Gain/switching: SPK_GAIN = GVDD_REG x 100k/250k = 2.8 V at GVDD 7.0 V -> 20 dBV, 8 x fSYNC = 768 kHz @48 kHz (Table 15, window 2.33-3.10 V holds for GVDD 5.83-7.75 V). D22: software ceiling -4.0 dBFS keeps 10 W into 4 Ohm.
LC: type 2 (L per leg, C to GND) as Fig. 79 and SLAA701B 1.3.2 for BD modulation; 10 uH + 0.68 uF: f0 61 kHz, Q 0.52 at 4 Ohm (-0.8 dB at 20 kHz). Power budget cited SLAA701B 1.3.4.1 = type 1 (differential C) - corrected.
CSBA0B50-100M: Isat 3.50 A (L -10 %) vs normal peak 2.6 A (10 W/4 Ohm 2.24 A + 0.33 A ripple at 768 kHz); at the 7.5 A OCE L >= 7 uH (-30 % at 8.00 A) vs 4.7 uH nominal / 1 uH min under short circuit (SLAS988B 7.3; CODACA 2026-03-16).
Caps: TI Table 22 values; 0402 parts moved to 0603 (hand assembly D42). PVDD caps > 1.45 x PVDD, LC caps > 1.8 x PVDD.
Overlay rpi-digiampplus: pcm5122@4c on i2c1, i2s_clk_producer, mute-gpios GPIO22 active high (driver sets 1 = unmute).
Thermal pad: via array + paste windows 70-80 % of the pad area (SLAS988B 11.1.3.2) - footprint in block 2.
No PVDD electrolytic at the amp (block 2, layout density): TI Table 22 says the 390 uF bulk "can be reduced or even removed"; C5 (120 uF on +20V) feeds PVDD through F5 (75 mOhm max, tau 9 us), plus 4 x 22 uF + 2 x 0.1 uF at the PVDD pins.

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Sheet: /Amplifier/
File: amplifier.kicad_sch

Title: Main board - amplifier

Size: A3	Date: 2026-11-07	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 5/5